$\nabla_t \times E(x,y) = 0$

$k_c^2 + \beta^2 = k^2$

TEM: $\frac{E_x}{H_y} = \eta$

Assignment sect.

$\frac{\partial}{\partial z} \to \beta^2$

$\nabla \times E_z = -j\omega\mu H$

$\nabla \times H = j\omega\epsilon E$

$$\nabla \times E = \frac{1}{\rho}\begin{vmatrix} \hat{\rho} & \rho\hat{\phi} & \hat{z} \\ \partial/\partial\rho & \partial/\partial\phi & \partial/\partial z \\ E_\rho & \rho E_\phi & E_z \end{vmatrix} = \left(\frac{\partial E_z}{\partial\phi} - \rho\frac{\partial E_\phi}{\partial z}\right)\hat{\rho} + \left(\frac{\partial E_\rho}{\partial z} - \frac{\partial E_z}{\partial\rho}\right)\rho\hat{\phi} + \left(E_\phi + \rho\frac{\partial E_\phi}{\partial\rho} - \frac{\partial E_\rho}{\partial\phi}\right)\hat{z}$$

$\frac{\partial}{\partial\rho}(\rho E_\phi) = E_\phi + \rho\frac{\partial E_\phi}{\partial\rho}$

$$\nabla \times E = \left(\frac{\partial E_z}{\partial\phi} + j\rho\beta E_\phi\right)\hat{\rho} - \left(\frac{\partial E_z}{\partial\rho} + j\beta E_\rho\right)\rho\hat{\phi} + \left(E_\phi + \rho\frac{\partial E_\phi}{\partial\rho} - \frac{\partial E_\rho}{\partial\phi}\right)\hat{z}$$

$$= -j\omega\mu\left(H_\rho\hat{\rho} + H_\phi\hat{\phi} + H_z\hat{z}\right)$$

① $\frac{\partial E_z}{\partial\phi} + j\rho\beta E_\phi = -j\omega\mu\rho H_\rho$

② $\rho\frac{\partial E_z}{\partial\rho} + j\beta\rho E_\rho = j\omega\mu\rho H_\phi$

③ $E_\phi + \rho\frac{\partial E_\phi}{\partial\rho} - \frac{\partial E_\rho}{\partial\phi} = -j\omega\mu\rho H_z$

④ $\frac{\partial H_z}{\partial\phi} + j\rho\beta H_\phi = j\omega\epsilon E_\rho\rho$

⑤ $\rho\frac{\partial H_z}{\partial\rho} + j\beta\rho H_\rho = -j\omega\epsilon E_\phi\rho$

⑥ $H_\phi + \rho\frac{\partial H_\phi}{\partial\rho} - \frac{\partial H_\rho}{\partial\phi} = j\omega\epsilon E_z\rho$

$$\nabla \times H = j\omega\epsilon E = \frac{1}{\rho}\begin{vmatrix} \hat{\rho} & \rho\hat{\phi} & \hat{z} \\ \partial/\partial\rho & \partial/\partial\phi & \partial/\partial z \\ H_\rho & \rho H_\phi & H_z \end{vmatrix} = \left(\frac{\partial H_z}{\partial\phi} + j\rho\beta H_\phi\right)\hat{\rho} - \left(\frac{\partial H_z}{\partial\rho} + j\beta H_\rho\right)\rho\hat{\phi}$$

$$+ \left(H_\phi + \rho\frac{\partial H_\phi}{\partial\rho} - \frac{\partial H_\rho}{\partial\phi}\right)\hat{z} = j\omega\epsilon\left(E_\rho\hat{\rho} + E_\phi\hat{\phi} + E_z\hat{z}\right)$$

① $\frac{\partial E_z}{\partial\phi} + j\rho\beta E_\phi = -j\omega\mu\rho H_\rho$ ⑤ $\rho\frac{\partial H_z}{\partial\rho} + j\beta\rho H_\rho = -j\omega\epsilon E_\phi\rho$ $\times j\rho\beta$

$\Rightarrow j\omega\epsilon\frac{\partial E_z}{\partial\phi} = \rho\beta\omega\epsilon E_\phi + \omega^2\mu\epsilon H_\rho$

② $\rho\frac{\partial E_z}{\partial\rho} + j\beta\rho E_\rho = j\omega\mu\rho H_\phi$ ④ $\frac{\partial H_z}{\partial\phi} + j\rho\beta H_\phi = j\omega\epsilon E_\rho\rho$ $\times -\beta$

$\Rightarrow j\rho^2\beta\frac{\partial H_z}{\partial\rho} = +\beta^2\rho^2 H_\rho + \rho\beta\omega\epsilon E_\phi$ $\ominus$

$$j\omega\epsilon\frac{\partial E_z}{\partial\phi} - j\rho^2\beta\frac{\partial H_z}{\partial\rho} = \omega^2\mu\epsilon H_\rho - \beta^2\rho^2 H_\rho = H_\rho\left(\omega^2\mu\epsilon - \beta^2\rho^2\right)$$

$$= H_\rho\left(k^2 - \beta^2\rho^2\right)$$

$$\begin{pmatrix} E_\rho \\ E_\phi \\ H_\rho \\ H_\phi \end{pmatrix} = \frac{j}{k_c^2}\begin{pmatrix} -\beta & 0 & 0 & \frac{-\omega\mu}{\rho} \\ 0 & \frac{-\beta}{\rho} & \omega\mu & 0 \\ 0 & \frac{\omega\epsilon}{\rho} & -\beta & 0 \\ -\omega\epsilon & 0 & 0 & \frac{-\beta}{\rho} \end{pmatrix}\begin{pmatrix} \partial E_z/\partial\rho \\ \partial E_z/\partial\phi \\ \partial H_z/\partial\rho \\ \partial H_z/\partial\phi \end{pmatrix}$$

① & ⑤ $\frac{\partial E_z}{\partial \phi} + j\rho\beta E_\phi = -j\omega\mu\rho H_\rho$ ① $\times j\omega\epsilon\rho$ ② $\times j\beta\rho$ $\Big/ \; j\omega\epsilon\rho\frac{\partial E_z}{\partial \phi} = \rho^2\beta\omega\epsilon E_\phi + \omega^2\epsilon\mu\rho^2 H_\rho$

$\rho\frac{\partial H_z}{\partial \rho} + j\beta\rho H_\rho = -j\omega\epsilon E_\phi\rho$ ① $\times j\rho\beta$ ② $\times j\omega\mu\rho$ $\Big/ \; j\rho^2\beta\frac{\partial H_z}{\partial \rho} = \beta^2\rho^2 H_\rho + \omega\epsilon\rho^2\beta E_\phi$ (−)

$= j\omega\epsilon\rho\frac{\partial E_z}{\partial \phi} - j\rho^2\beta\frac{\partial H_z}{\partial \rho} = \omega^2\mu\epsilon\rho^2 H_\rho - \beta^2\rho^2 H_\rho = H_\rho(\omega^2\mu\epsilon\rho^2 - \beta^2\rho^2)$

$= H_\rho(+k_c^2\rho^2)$

① $H_\rho = \frac{j}{\rho^2 k_c^2}\left(\omega\epsilon\rho\frac{\partial E_z}{\partial \phi} - \rho\beta\frac{\partial H_z}{\partial \rho}\right)$

$j\beta\rho\frac{\partial E_z}{\partial \phi} = \rho^2\beta^2 E_\phi + \omega^2\mu\epsilon\rho^2 H_\rho$ (−)

$j\omega\mu\rho^2\frac{\partial H_z}{\partial \rho} = \beta\rho^2\omega\mu H_\rho + \omega^2\mu\epsilon\rho^2 E_\phi$

$j\beta\rho\frac{\partial E_z}{\partial \phi} - j\omega\mu\rho^2\frac{\partial H_z}{\partial \rho} = \rho^2\beta^2 E_\phi - \omega^2\mu\epsilon\rho^2 E_\phi = E_\phi(\rho^2\beta^2 - k^2\rho^2)$

$= E_\phi(-k_c^2\rho^2)$

② $E_\phi = \frac{-j}{\rho^2 k_c^2}\left(\beta\rho\frac{\partial E_z}{\partial \phi} - \omega\mu\rho^2\frac{\partial H_z}{\partial \rho}\right)$

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② & ④ $\rho\frac{\partial E_z}{\partial \rho} + j\beta\rho E_\rho = j\omega\mu\rho H_\phi$ $\times j\omega\epsilon\rho$ $\times j\beta$ $\quad j\omega\epsilon\rho^2\frac{\partial E_z}{\partial \rho} = \beta\rho^2\omega\epsilon E_\rho - \omega^2\mu\epsilon\rho^2 H_\phi$

$\frac{\partial H_z}{\partial \phi} + j\rho\beta H_\phi = j\omega\epsilon E_\rho\rho$ $\times j\beta\rho$ $\times j\omega\mu\rho$ $\quad j\beta\rho\frac{\partial H_z}{\partial \phi} = \rho^2\beta^2 H_\phi - \omega\epsilon\beta\rho^2 E_\rho$ (−)

$j\omega\epsilon\rho^2\frac{\partial E_z}{\partial \rho} + j\beta\rho\frac{\partial H_z}{\partial \phi} = -\omega^2\mu\epsilon\rho^2 H_\phi + \rho^2\beta^2 H_\phi = H_\phi(\rho^2\beta^2 - \omega^2\mu\epsilon\rho^2)$ ($\omega^2\mu\epsilon = k^2$)

$= H_\phi(\rho^2(-k_c^2))$

③ $H_\phi = \frac{-j}{\rho^2 k_c^2}\left(\omega\epsilon\rho^2\frac{\partial E_z}{\partial \rho} + \beta\rho\frac{\partial H_z}{\partial \phi}\right)$

$j\rho^2\beta\frac{\partial E_z}{\partial \rho} = \beta^2\rho^2 E_\rho - \omega\mu\rho^2\beta H_\phi$

$j\omega\mu\rho\frac{\partial H_z}{\partial \phi} = \rho^2\omega\mu\beta H_\phi - \omega^2\mu\epsilon\rho^2 E_\rho$ (+)

$j\rho^2\beta\frac{\partial E_z}{\partial \rho} + j\omega\mu\rho\frac{\partial H_z}{\partial \phi} = \beta^2\rho^2 E_\rho + -\omega^2\mu\epsilon\rho^2 E_\rho = E_\rho(\beta^2\rho^2 - \omega^2\mu\epsilon\rho^2)$

$= E_\rho(\rho^2(-k_c^2))$

④ $E_\rho = \frac{-j}{\rho^2 k_c^2}\left(\rho\rho^2\beta\frac{\partial E_z}{\partial \rho} + \omega\mu\rho\frac{\partial H_z}{\partial \phi}\right)$

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① $E_\rho = \frac{-j}{k_c^2}\left(\beta\frac{\partial E_z}{\partial \rho} + \frac{\omega\mu}{\rho}\frac{\partial H_z}{\partial \phi}\right)$ ② $E_\phi = \frac{-j}{k_c^2}\left(\frac{\beta}{\rho}\frac{\partial E_z}{\partial \phi} - \omega\mu\frac{\partial H_z}{\partial \rho}\right)$

③ $H_\rho = \frac{j}{k_c^2}\left(\frac{\omega\epsilon}{\rho}\frac{\partial E_z}{\partial \phi} - \beta\frac{\partial H_z}{\partial \rho}\right)$ ④ $H_\phi = \frac{-j}{k_c^2}\left(\omega\epsilon\frac{\partial E_z}{\partial \rho} + \frac{\beta}{\rho}\frac{\partial H_z}{\partial \phi}\right)$

$$\nabla\times\vec{E} = \frac{1}{\rho}\begin{vmatrix} \hat{\rho} & \rho\hat{\phi} & \hat{z} \\ \partial/\partial\rho & \partial/\partial\phi & \partial/\partial z \\ E_\rho & \rho E_\phi & E_z \end{vmatrix} = \hat{\rho}\left(\frac{1}{\rho}\frac{\partial E_z}{\partial\phi} - \frac{\partial E_\phi}{\partial z}\right) + \hat{\phi}\left[\frac{\partial E_\rho}{\partial z} - \frac{\partial E_z}{\partial\rho}\right] + \hat{z}\left[\frac{\partial E_\phi}{\partial\rho} + \frac{E_\phi}{\rho} - \frac{\partial E_\rho}{\rho\partial\phi}\right]$$

$$\nabla\times\nabla\times\vec{E} = \frac{1}{\rho}\begin{vmatrix} \hat{\rho} & \rho\hat{\phi} & \hat{z} \\ \partial/\partial\rho & \partial/\partial\phi & \partial/\partial z \\ A & B & C \end{vmatrix} =$$

$$= \hat{\rho}\left(\frac{1}{\rho}\frac{\partial C}{\partial\phi} - \frac{\partial B}{\partial z}\right) + \hat{\phi}\left(\frac{\partial A}{\partial z} - \frac{\partial C}{\partial\rho}\right) + \hat{z}\left[\frac{1}{\rho}\frac{\partial(\rho B)}{\partial\rho} - \frac{1}{\rho}\frac{\partial A}{\partial\phi}\right] =$$

$$\frac{\partial E_\rho}{\partial z_1} = \frac{1}{\rho}\left[\frac{\partial z_2}{\partial\phi}\frac{\partial E}{\partial z_2} - \frac{\partial\rho_2}{\partial\phi}\frac{\partial E}{\partial\rho_2} + \frac{\partial\phi}{\partial\phi}\frac{\partial E}{\partial\phi}\right]$$

$$\frac{\partial E}{\partial B} = \frac{\partial E}{\partial z_2}\frac{\partial z_2}{\partial\rho} - \frac{\partial E}{\partial\rho_2}$$

$$\frac{\partial E}{\partial A} = \frac{\partial E}{\partial z_2}\frac{\partial z_2}{\partial\phi}\frac{1}{\rho} + \frac{\partial E}{\partial\phi}\frac{\partial\phi}{\partial z}$$

$$\frac{\partial E}{\partial C} = \frac{\partial E}{\partial\phi}\frac{\partial\phi}{\partial z} + \frac{\partial E}{\partial z_2}\frac{\partial z_2}{\partial z} + \frac{\partial E}{\partial\rho_2}\frac{\partial\rho_2}{\partial z}$$

$$\frac{\partial E}{\partial(\rho B)}\frac{1}{\rho} = \frac{\partial E}{\partial z_2}\frac{\partial z_2}{\partial\rho}\frac{1}{\rho} + \frac{\partial E}{\partial\rho_2}\frac{\partial\rho_2}{\partial\rho}\frac{1}{\rho}$$

$$\frac{\partial E}{\partial A}\frac{1}{\rho} = \frac{\partial E}{\partial\rho_2}\frac{\partial\rho_2}{\partial\phi}\frac{1}{\rho} - \frac{\partial E}{\partial z_2}\frac{\partial z_2}{\partial\phi}\frac{1}{\rho}$$

$$\nabla\times\nabla\times\vec{E} = \hat{\rho}\left[\frac{1}{\rho}\left(\frac{\partial E}{\partial z_2}\frac{\partial z_2}{\partial\phi}\frac{1}{\rho} + \frac{\partial E}{\partial\rho_2}\frac{\partial\rho_2}{\partial\phi}\frac{1}{\rho} + \frac{\partial E}{\partial\phi}\frac{1}{\rho}\right) - \left[\frac{\partial E}{\partial z_2}\frac{\partial z_2}{\partial\rho} + \frac{\partial E}{\partial\rho_2}\frac{\partial\rho_2}{\partial\rho}\right]\right] +$$

$$+\hat{\phi}\left[\left[\frac{\partial E}{\partial z_2}\frac{\partial z_2}{\partial z} + \frac{\partial E}{\partial\rho_2}\frac{\partial\rho_2}{\partial z} - \frac{\partial E}{\partial z}\right] - \left[\frac{\partial E}{\partial z_2}\frac{\partial z_2}{\partial\rho} + \frac{\partial E}{\partial\rho_2}\frac{\partial\rho_2}{\partial\rho} - \frac{1}{\rho}\frac{\partial E}{\partial\phi}\right]\right] +$$

$$+\hat{z}\left[\frac{\partial E}{\partial\rho_2}\frac{\partial\rho_2}{\partial\phi}\frac{1}{\rho} + \frac{\partial E}{\partial z_2}\frac{\partial z_2}{\partial\phi}\frac{1}{\rho} - \frac{1}{\rho}\frac{\partial E}{\partial\phi} - \frac{\partial E}{\partial z_2}\frac{\partial z_2}{\partial z} + \frac{1}{\rho}\frac{\partial E}{\partial z}\right]$$

$$\nabla\cdot\vec{E} = \frac{\partial E}{\partial z_2}\frac{\partial z_2}{\partial z} + \frac{\partial E}{\partial\phi}\frac{\partial\phi}{\partial z} + \frac{\partial E}{\partial\rho_2}\frac{\partial\rho_2}{\partial z} + \frac{E}{\rho}\frac{\partial}{\partial\rho}(\rho E_\rho) =$$

$$= \nabla\cdot\vec{A} = \frac{\partial E}{\partial z_2}\frac{\partial z_2}{\partial z} + \frac{\partial E}{\partial\phi}\frac{\partial\phi}{\partial z} + \frac{\partial E}{\partial\rho_2}\frac{\partial\rho_2}{\partial z}$$

$$(\nabla\cdot\vec{A})\hat{\rho} = \left[\frac{\partial E}{\partial\rho_2}\frac{\partial\rho_2}{\partial\phi}\frac{1}{\rho} + \frac{\partial E}{\partial\phi}\frac{\partial\phi}{\partial\rho}\frac{1}{\rho} - \frac{1}{\rho}\frac{\partial}{\partial\rho}(\rho E_\rho) + \frac{\partial E}{\partial z_2}\frac{\partial z_2}{\partial\rho}\right]\hat{\rho} +$$

$$+\left(\frac{\partial E}{\partial z_2}\frac{\partial z_2}{\partial\phi}\frac{1}{\rho} + \frac{\partial E}{\partial\phi}\frac{\partial\phi}{\partial\phi}\frac{1}{\rho} + \frac{\partial E}{\partial\rho_2}\frac{\partial\rho_2}{\partial\phi}\frac{1}{\rho}\right)\hat{\phi} +$$

$$+\left(\frac{\partial E}{\partial z_2}\frac{\partial z_2}{\partial z} + \frac{\partial E}{\partial\phi}\frac{\partial\phi}{\partial z} + \frac{\partial E}{\partial\rho_2}\frac{\partial\rho_2}{\partial z}\right)\hat{z}$$

$$\nabla^2 E = \nabla(\nabla \cdot E) - \nabla \times \nabla \times \vec{E}$$

comparing $\hat{z}$ as sample.

$$\frac{\partial^2 E_\rho}{\partial \rho \partial z} + \frac{1}{\rho}\frac{\partial E_\rho}{\partial z} + \frac{1}{\rho}\frac{\partial^2 E_\phi}{\partial \phi \partial z} + \frac{\partial^2 E_z}{\partial z^2} - \left[\frac{\partial E_\rho}{\partial z}\frac{1}{\rho} + \frac{\partial^2 E_\rho}{\partial \phi \partial z}\frac{1}{\rho} - \frac{\partial^2 E_z}{\partial \rho \partial z} - \frac{1}{\rho^2}\frac{\partial^2 E_z}{\partial \phi^2}\right]$$

$$= \frac{1}{\rho}\frac{\partial^2 E_\rho}{\partial \phi \partial z} + \frac{\partial E_z}{\partial \rho}\frac{1}{\rho} + \frac{\partial^2 E_z}{\partial \rho^2} + \frac{1}{\rho^2}\frac{\partial^2 E_z}{\partial \phi^2}$$

$$\nabla^2 = \frac{\partial^2}{\partial \rho^2} + \frac{1}{\rho}\frac{\partial}{\partial \rho} + \frac{1}{\rho^2}\frac{\partial^2}{\partial \phi^2} + \frac{\partial^2}{\partial z^2} \Rightarrow \nabla^2 E = \left[\frac{\partial^2 E_\rho}{\partial \rho^2} + \frac{1}{\rho}\frac{\partial E_\rho}{\partial \rho} + \frac{1}{\rho^2}\frac{\partial^2 E_\rho}{\partial \phi^2} + \frac{\partial^2 E_\rho}{\partial z^2}\right]\hat{\rho}$$

$$\nabla^2 E = \hookrightarrow \left(\frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho \frac{\partial E_z}{\partial \rho}\right) + \frac{1}{\rho^2}\frac{\partial^2 E_z}{\partial \phi^2} + \frac{\partial^2 E_z}{\partial z^2}\right)\hat{z}$$

$$(\nabla^2 E)_z = \frac{1}{\rho}\frac{\partial E_z}{\partial \rho} + \frac{\partial^2 E_z}{\partial \rho^2} + \frac{1}{\rho^2}\frac{\partial^2 E_z}{\partial \phi^2} + \frac{\partial^2 E_z}{\partial z^2}$$

comparing we find 2 terms that aren't equal

Sheet (3)

Problem (13)

$$\beta = \sqrt{\frac{\omega^2}{c^2} - \frac{\omega_c^2}{c^2}} = \frac{1}{c}\sqrt{\omega^2 - \omega_c^2}$$

$$\frac{\partial \beta}{\partial \omega} = \frac{1}{c}\frac{\omega}{(\omega^2 - \omega_c^2)^{1/2}} \Rightarrow \frac{\partial^2 \beta}{\partial \omega^2} = \frac{1}{c}\frac{(\omega^2 - \omega_c^2)^{1/2} - \omega\frac{\omega}{\sqrt{\omega^2 - \omega_c^2}}}{(\omega^2 - \omega_c^2)}$$

$$\frac{\partial^2 \beta}{\partial \omega^2} = \frac{1}{c}\left[(\omega^2 - \omega_c^2)^{-1/2} - \omega^2(\omega^2 - \omega_c^2)^{-3/2}\right]$$

$$= \frac{1}{c}\left(\frac{1}{\sqrt{\omega^2 - \omega_c^2}} - \omega^2\frac{1}{(\omega^2 - \omega_c^2)^{3/2}}\right) = -\frac{1}{c}\left(\frac{1}{\omega}\frac{1}{(1 - \frac{\omega_c^2}{\omega^2})^{3/2}} - \frac{1}{\omega(1 - \frac{\omega_c^2}{\omega^2})^{1/2}}\right)$$

$$= -\frac{1}{\omega c}\frac{1}{(1 - \frac{\omega_c^2}{\omega^2})^{3/2}}\left[1 - \left[1 - \frac{\omega_c^2}{\omega^2}\right]\right] = -\frac{1}{\omega c}\frac{\omega_c^2}{\omega^2}\left[1 - \frac{\omega_c^2}{\omega^2}\right]^{-3/2}$$